Fiscal Year:	FY 2018	**Task Last Updated:**	FY 06/01/2018
PI Name:	Urbaniak, Camilla Ph.D.		
Project Title:	Characterization of the Environmental Microbiome of the International Space Station		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
PI Email:	camilla.urbaniak@jpl.nasa.gov	**Fax:**	FY
PI Organization Type:	INDUSTRY	**Phone:**	626-840-6992
Organization Name:	ZIN Technologies, Inc		
PI Address 1:	4800 Oak Grove Dr		
PI Address 2:			
PI Web Page:			
City:	Pasadena	**State:**	CA
Zip Code:	91109-8001	**Congressional District:**	27
Comments:	NOTE: PI had NASA Postdoctoral Program Fellowship at NASA Jet Propulsion Laboratory until 2019; now with Zin Technologies, Inc., Middleburg, OH, but remains physically at NASA JPL address.		
Project Type:	Ground	**Solicitation / Funding Source:**	NASA Postdoctoral Program
Start Date:	08/01/2016	**End Date:**	07/31/2019
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JPL
Contact Monitor:	Administration, USRA	**Contact Phone:**	
Contact Email:	npphelp@usra.edu		
Flight Program:			
Flight Assignment:	NOTE: Extended to 7/31/2019 per JPL (Ed., 12/31/18)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Venkateswaran, Kasthuri Ph.D. (MENTOR/ NASA Jet Propulsion Laboratory)		
Grant/Contract No.:	NASA Postdoctoral Program		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NASA Postdoctoral Program Fellow Using next generation sequencing and microarrays, the microbial communities present on the International Space Station (ISS), over a period of one year, will be characterized. Factors affecting this microbial composition and those that promote pathogen transmission between surfaces and the crew will be assessed. In addition, the effects of spaceflight on antibiotic resistance and virulence factors will be examined. Findings from this study will allow mission planners to design better strategies to mitigate health risk to astronauts while in Space.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	• lead author in a recently published paper analyzing the collection of anti-microbial genes (“resistome”) found on the ISS using a variety of molecular techniques. • performed whole genome sequencing on 4 fungal isolates (Fusarium oxysporum) isolated from the ISS during MT-1 and VEGGIE projects. Performed comparative genomics and single nucleotide polymorphism (SNP) analysis on these genomes and believes that at least two may be a novel strain. • analyzed the microbial load and microbial diversity of ISS surface samples collected during Microbial Tracking-1 and is a co-author on a paper that is under review by mSphere. • provided invaluable help to colleagues in processing samples from a variety of Jet Propulsion Laboratory (JPL) and NASA collaborative projects. • is a Co-Investigator (Co-I) and helped extensively in the write up of the recently awarded grant: “Omics in Space (OIS): Technology Development for Omics Instrumentations and Biomolecule Measurement” • developed a project here at JPL to assess the impact of microgravity on the rate of horizontal gene transfer between different bacterial genera. Preliminary results are very interesting and more work is being conducted. • wrote and submitted a ROSBio grant (as a Co-I) to help support work on the impact of microgravity on anti-microbial resistance and virulence. • presented her research in oral presentations at American Society for Gravitational and Space Research-ASGSR (Cleveland and Seattle), AbSciCon (Mesa, Arizona), an International Astronautical Congress (Australia) meetings. • Postdoc strives to keep her skills current and took a week-long course in RNA-seq analysis. • MT-2 project: has processed: (a) SpaceX Dragon surface and air samples, (b) processed ISS surface and air samples collected from two flight sampling events, (c) performed culture and molecular analysis to measure microbial burden, (d) archived DNA and microbial isolates, and (e) coordinated with collaborating institutions like Lawrence Livermore, Johnson Space Center-JSC, and NASA Ames for sample analyses. She prepared debriefings on numerous occasions for all those involved in the project. In addition to the environmental samples, she has processed crew samples collected from two subjects to date.
Bibliography Type:	Description: (Last Updated: 11/01/2024)
Abstracts for Journals and Proceedings	Grams T, Urbaniak C, Henle A, Venkateswaran K. "Assessing the ability of a microgravity environment to promote the transfer of antibiotic resistance." 33rd Annual Meeting of the American Society for Gravitational and Space Research, Seattle, WA, October 25-28, 2017. 33rd Annual Meeting of the American Society for Gravitational and Space Research, Seattle, WA, October 25-28, 2017. , Oct-2017
Articles in Peer-reviewed Journals	Urbaniak C, Dadwal S, Bagramyan K, Venkateswaran K. "Draft genome sequence of a clinical isolate of Fusarium fujikuroi isolated from a male patient with acute myeloid leukemia." Genome Announc. 2018 Jun 21;6(25):e00476-18. https://doi.org/10.1128/genomeA.00476-18 ; PubMed PMID: 29930033 ; PubMed Central PMCID: PMC6013642 , Jun-2018
Articles in Peer-reviewed Journals	Urbaniak C, Massa G, Hummerick M, Khodadad C, Schuerger A, Venkateswaran K. "Draft genome sequences of two Fusarium oxysporum isolates cultured from infected Zinnia hybrida plants grown on the International Space Station. " Genome Announc. 2018 May 17;6(20):e00326-18. https://doi.org/10.1128/genomeA.00326-18 ; PubMed PMID: 29773617 ; PubMed Central PMCID: PMC5958250 , May-2018
Articles in Peer-reviewed Journals	Urbaniak C, Sielaff AC, Frey KG, Allen JE, Singh N, Jaing C, Wheeler K, Venkateswaran K. "Detection of antimicrobial resistance genes associated with the International Space Station environmental surfaces." Sci Rep. 2018 Jan 16;8(1):814. https://doi.org/10.1038/s41598-017-18506-4 ; PubMed PMID: 29339831; PubMed Central PMCID: PMC5770469 , Jan-2018
Papers from Meeting Proceedings	Hendrickson R, Lundgren P, Malli Mohan GB, Urbaniak C, Benardini JN, Venkateswaran K. "Comprehensive measurement of microbial burden in nutrient-deprived cleanrooms." 47th International Conference on Environmental Systems, Charleston, SC, July 16-20, 2017. 47th International Conference on Environmental Systems, Charleston, SC, July 16-20, 2017. ICES paper ICES-2017-177. , Jul-2017